ATI COURSE PROJECT 2026 · TRACK 2 · MIDTERM REPORT

ViGoEmotions: Fine-Grained Multi-Label Emotion Recognition for Vietnamese Social-Media Comments

Dataset analysis, fixed split, preprocessing, baseline models, evaluation, error analysis and the proposed Final research question

Team: [Team name] — [Member 1], [Member 2], [Member 3]

Date: [Submission date]

Code: executable notebooks and Python package in this submission (README.md); all tables and figures are regenerated by code.

Abstract

We study fine-grained multi-label emotion recognition on ViGoEmotions (EACL 2026), a corpus of 20,664 Vietnamese social-media comments annotated with 27 emotions plus neutral. We (i) analyse the label space, co-occurrence structure and informal-language phenomena; (ii) keep the official 80/10/10 split and audit its multi-label stratification; (iii) build a documented Vietnamese normalisation pipeline (Unicode NFC, VinAI tone normalisation, teencode and emoji handling, VnCoreNLP word segmentation); and (iv) benchmark the three required models — TF-IDF with One-vs-Rest Logistic Regression, a BiLSTM and PhoBERT with an independent sigmoid head trained with BCE — plus ViSoBERT, ViSoBERT-14GB and a probability-averaging ensemble, all evaluated with one shared module. At the default threshold $\tau = 0.5$, PhoBERT reaches **0.629 Macro-F1 / 0.646 Micro-F1** on the held-out test set, above the best published baseline on the same split (ViSoBERT, 0.615). The ensemble reaches 0.643 / 0.664, and 0.653 / 0.669 with a single threshold fitted on validation. Label frequency correlates only weakly with per-label F1, PhoBERT's largest gains over the BiLSTM are on emoticon, long and teencode comments, and validation-optimal per-label thresholds range from 0.17 to 0.69 — motivating the Final research question on threshold calibration.

1. Introduction

A single social-media comment frequently expresses several emotions at once (for example sadness together with disappointment). The task is therefore multi-label: each of the 28 categories is an independent binary decision, and the model outputs one sigmoid probability per label instead of a softmax over mutually exclusive classes. Following the project brief, Macro-F1 and Micro-F1 are the headline metrics; exact-match subset accuracy is reported for completeness only.

Vietnamese social-media text adds difficulties absent from edited text: teencode and abbreviations (“ko”, “đc”, “vs”), elongated words, emoji and ASCII emoticons (“:))”, “=))”), missing diacritics and profanity. This Midterm establishes a clean, reproducible baseline study; the contributions are:

- a complete dataset analysis (inventory, frequency tiers, cardinality, co-occurrence/correlation, text length and an informal-language audit), computed from code;
- a fixed split with a quantitative stratification audit and frozen text-style subset definitions;
- a Vietnamese preprocessing pipeline that fixes a Unicode-normalisation bug in the official code and aligns tokenisation with each encoder's pre-training;
- the three required models plus stronger Vietnamese encoders and an ensemble, evaluated with all required metrics, parameter counts and wall-clock time;
- evidence-based answers to the five Midterm questions, an error analysis and the proposed Final research question.

2. Dataset and access protocol

ViGoEmotions (Hung et al., EACL 2026) contains 20,664 comments collected from Facebook (including the earlier UIT-VSMEC corpus), TikTok, YouTube, X, Reddit and Threads. The dataset is distributed as a gated Hugging Face dataset (uitnlp/vigoemotions) and in the authors' repository under a non-commercial research licence that forbids redistribution. Our submission therefore contains code, split identifiers, aggregate statistics, labels and predicted probabilities, but no raw comments; the manual error audit that needs text is written to a git-ignored private folder.

Table 1. Dataset summary (computed by `python -m vigo eda`).

Item	Value
total comments	20,664
train comments	16,531
val comments	2,066
test comments	2,067
labels (27 emotions + neutral)	28
mean labels per comment |Y|	1.909
label density	0.0682
distinct label sets	1,485
most frequent label (train)	amusement
rarest label (train)	relief
imbalance ratio max/min (train)	4.52
median whitespace tokens (train)	11

Table 2. Source platform inferred from the sample-id prefix.

Source	Train	Val	Test	Total	Share
Facebook (numeric id, incl. UIT-VSMEC)	10,741	1,339	1,364	13,444	65.1%
TikTok	3,609	463	436	4,508	21.8%
YouTube	1,563	199	198	1,960	9.5%
X (Twitter)	430	48	46	524	2.5%
Reddit	140	13	14	167	0.8%
Threads	48	4	9	61	0.3%

Table 3. Complete label inventory with positives per split and the frequency tier (training counts).

#	Label	Train	Val	Test	Train %	Tier
0	amusement	2,868	327	374	17.4%	head
1	excitement	824	91	98	5.0%	mid
2	joy	1,614	168	204	9.8%	head
3	love	1,175	148	143	7.1%	head
4	desire	762	102	80	4.6%	tail
5	optimism	1,046	122	142	6.3%	mid
6	caring	1,062	131	150	6.4%	head
7	pride	774	87	86	4.7%	mid
8	admiration	883	107	101	5.3%	mid
9	gratitude	765	96	108	4.6%	mid
10	relief	635	58	60	3.8%	tail
11	approval	917	119	115	5.5%	mid
12	realization	687	96	95	4.2%	tail
13	surprise	651	87	85	3.9%	tail
14	curiosity	840	112	100	5.1%	mid
15	confusion	676	89	84	4.1%	tail
16	fear	745	117	98	4.5%	tail
17	nervousness	728	109	96	4.4%	tail
18	remorse	707	96	87	4.3%	tail

#	Label	Train	Val	Test	Train %	Tier
19	embarrassment	885	137	107	5.3%	mid
20	disappointment	1,522	211	197	9.2%	head
21	sadness	2,785	354	345	16.9%	head
22	grief	747	97	86	4.5%	tail
23	disgust	1,206	164	141	7.3%	head
24	anger	1,589	185	189	9.6%	head
25	annoyance	2,662	331	343	16.1%	head
26	disapproval	969	123	112	5.9%	mid
27	neutral	821	94	116	5.0%	mid

3. Exploratory data analysis

3.1 Label frequency and head / mid / tail tiers

Positive counts range from 2,868 (amusement) to 635 (relief) in the training set, an imbalance ratio of only 4.5× — much milder than English GoEmotions. The tier rule was fixed before any evaluation: labels are ranked by training positive count and split at the tertiles, giving 9 head, 10 mid and 9 tail labels (Table 4). The same rule will be reused in the Final.

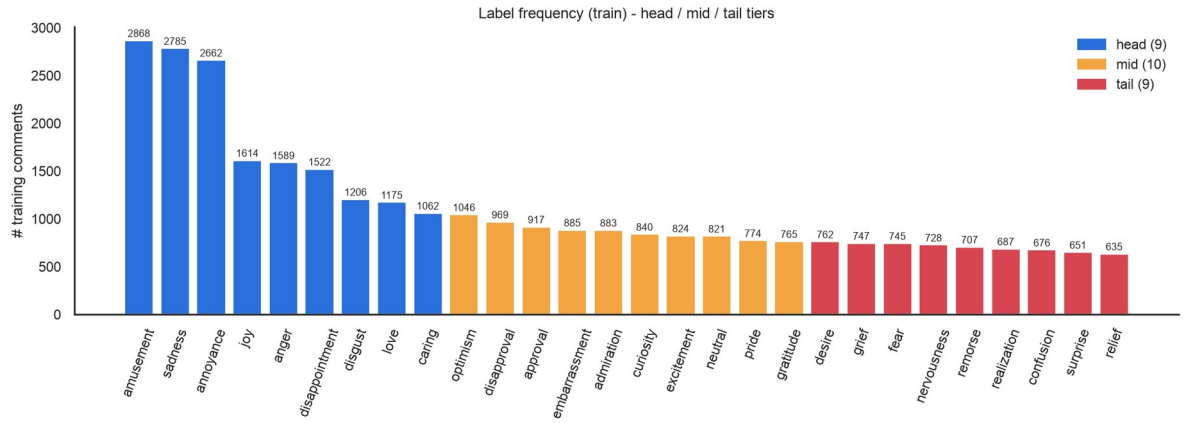


Figure 1. Training positives per label, coloured by tier.

Table 4. Frequency tiers (tertiles of training positive counts).

Tier	# labels	Min	Max	Labels
head	9	1,062	2,868	amusement, sadness, annoyance, joy, anger, disappointment, disgust, love, caring
mid	10	765	1,046	optimism, disapproval, approval, embarrassment, admiration, curiosity, excitement, neutral, pride, gratitude
tail	9	635	762	desire, grief, fear, nervousness, remorse, realization, confusion, surprise, relief

3.2 Label cardinality

Comments carry 1.91 labels on average (label density 0.068); 32.9% of training comments have exactly one label, 46.6% two, 17.6% three and 2.9% four or more. Only 3.5% are neutral-only, and 1,485 distinct label combinations occur. The distribution is almost identical across splits (Table 5), and |Y| is kept per sample for the cardinality-stratified analysis of the Final.

Table 5. Label cardinality per split (% of comments).

Split	n	Mean |Y|	|Y|=1	|Y|=2	|Y|=3	|Y|=4	|Y|=5	Neutral only
train	16,531	1.908	32.9%	46.6%	17.6%	2.6%	0.3%	3.5%
val	2,066	1.916	32.9%	45.7%	18.8%	2.3%	0.3%	3.6%
test	2,067	1.907	33.6%	45.8%	17.4%	2.7%	0.5%	3.9%

3.3 Co-occurrence and pairwise correlation

All co-occurrence statistics are computed on training labels only, following the reproducibility rules. The strongest pairs form interpretable emotion families: a sadness family (sadness, grief, disappointment, embarrassment, remorse), an anxiety pair (fear–nervousness, the highest phi), a positive cluster (excitement, joy, love, optimism, caring, gratitude–relief) and a hostile cluster (anger, annoyance, disapproval, disgust). Grief almost always co-occurs with sadness (P(sadness | grief) = 0.83). Amusement is negatively correlated with sadness.

Table 6. Most frequent label pairs in the training set.

Label a	Label b	Count	P(b|a)	P(a|b)	phi
disappointment	sadness	904	0.594	0.325	0.362
anger	annoyance	699	0.440	0.263	0.247
sadness	grief	620	0.223	0.830	0.384
amusement	joy	568	0.198	0.352	0.155
annoyance	disapproval	469	0.176	0.484	0.219
joy	love	395	0.245	0.336	0.222
disgust	anger	379	0.314	0.239	0.208
embarrassment	sadness	366	0.414	0.131	0.156
excitement	joy	359	0.436	0.222	0.261
disappointment	annoyance	336	0.221	0.126	0.052
amusement	annoyance	336	0.117	0.126	-0.055
fear	nervousness	322	0.432	0.442	0.411

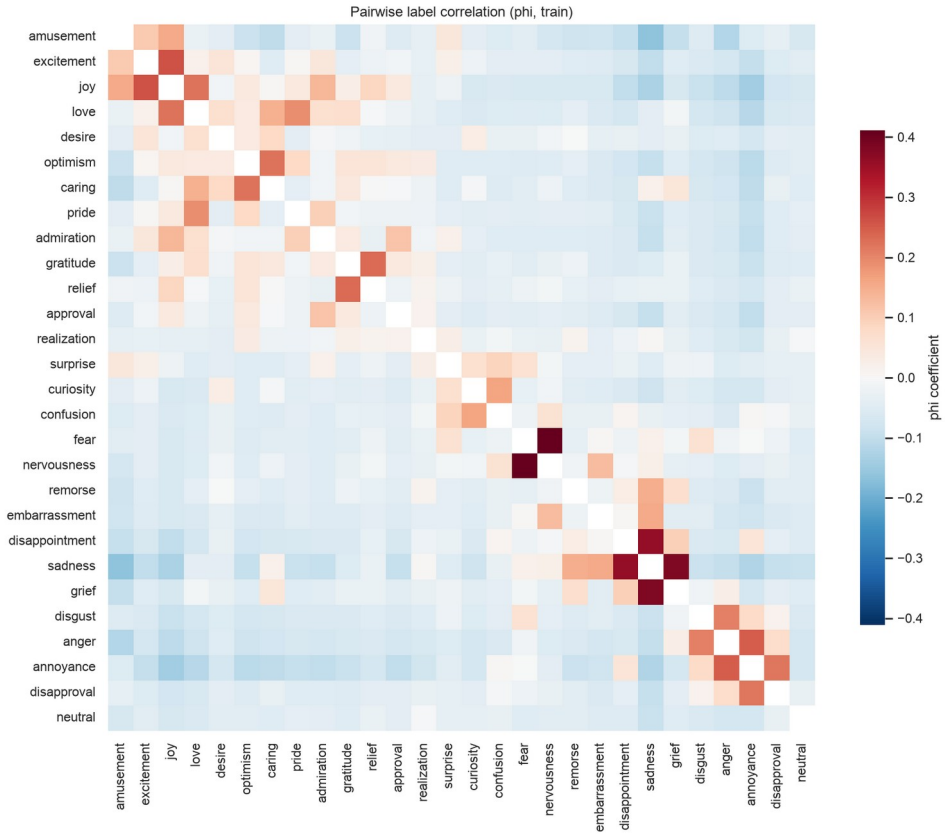


Figure 2. Pairwise phi correlation between labels (training set).

3.4 Text length

Comments are short: the median training comment has 11 whitespace tokens (47 characters), the 95th percentile 44 tokens and the 99th percentile 84 tokens. After PhoBERT tokenisation the 99th percentile is 85 sub-words, so max_len = 192 truncates only 0.01% of training comments; the BiLSTM uses 100 word tokens (above the 99th percentile of 78 segmented tokens).

Table 7. Comment length statistics.

Split	Unit	Min	P25	Median	Mean	P90	P95	P99	Max
train	tokens	1	7	11	16	32	44	84	214
train	characters	1	28	47	66.5	133	186	355	906
val	tokens	1	7	12	16.1	32	44	88	143
val	characters	1	27	47	66.5	130.5	187.8	363.4	641
test	tokens	1	7	11	16.8	35	47	90.7	171
test	characters	1	28	47	69.6	145	194	380	694

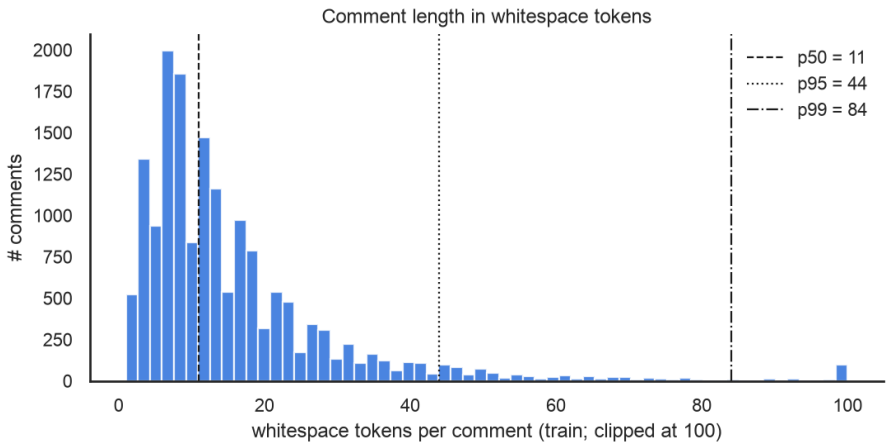


Figure 3. Distribution of whitespace tokens per training comment.

3.5 Informal Vietnamese: slang, teencode, emoji and non-standard syntax

Table 8 quantifies informal phenomena over all 20,664 comments (aggregates only). About a third of the comments contain at least one teencode or abbreviation token, a quarter contain emoji and one in nine contains an ASCII emoticon. Emoji are strongly tied to specific emotions: 48.8% of love comments and 44.4% of pride comments contain an emoji. These observations drive the preprocessing decisions: emoji are never deleted, teencode is expanded with a token-level dictionary, and elongations are squeezed.

Table 8. Informal-language audit (share of all comments).

Phenomenon	# comments	%
contains emoji	5,153	24.9%
>= 2 emoji	1,980	9.6%
ASCII emoticon (:)) =)) :v ^^)	2,385	11.5%
teencode / abbreviation token (dictionary)	6,559	31.7%
multi-char teencode token	4,862	23.5%
elongated word (letter x3+, e.g. 'quáaaa')	432	2.1%
repeated punctuation (!!, ??, ...)	1,535	7.4%
laughter token (haha, hihi, kkk, =))	684	3.3%
@mention	727	3.5%
URL	0	0.0%
hashtag	29	0.1%
ALL-CAPS word (>= 3 letters)	291	1.4%
unaccented Vietnamese (>= 4 tokens, no diacritics)	139	0.7%
anonymised person placeholder 'per'	631	3.0%
  / line break	614	3.0%
profanity / vulgar slang (dm, vl, vcl, đm, cc...)	1,048	5.1%

Table 9. Most frequent teencode tokens and their normalisation.

Token	Normalised	Count
ko	không	1,618
t	tao	1,096
k	không	965

Token	Normalised	Count
e	em	695
đc	được	598
a	anh	487
ng	người	404
v	vậy	299
r	rồi	290
c	chị	259
m	mày	220
j	gì	211

4. Train / validation / test protocol

We keep the official 80/10/10 split released with the dataset (16,531 / 2,066 / 2,067 comments) so that every number is directly comparable with the published baselines; the ids are stored in splits/official/. Because the brief asks for a stratified multi-label protocol, the "python -m vigo split" command audits this split against a random split and an iterative multi-label stratified split (Sechidis et al., 2011) generated with seed 42, using the label-distribution (LD) measure. The official split is only random-level stratified (mean LD 0.0035 vs 0.0044), whereas iterative stratification is about nine times better (0.0004); the rarest label (relief) has 58 validation positives. We retain the official split for comparability and store the iterative split in splits/iterative/ for a robustness check in the Final.

Table 10. Stratification audit of three 80/10/10 protocols (lower LD is better).

Protocol	Mean LD	Val LD	Test LD	Min val pos.	Min test pos.
official (released)	0.00355	0.00573	0.00415	58	60
random (seed 42)	0.00441	0.00469	0.00729	73	88
iterative stratified (seed 42)	0.00041	0.00067	0.00055	75	76

Discipline: the training set fits models, vocabularies and co-occurrence statistics; the validation set is used for early stopping, hyper-parameter selection and threshold fitting; the test set is scored once per frozen model. The analysis script asserts that every model's test predictions are aligned with the same ordered test ids.

5. Preprocessing pipeline

The pipeline (src/vigo/data/preprocessing.py) is configurable per model and cached by a hash of the configuration and the code. Steps, in order:

1. **Unicode NFC** and removal of emoji variation selectors and skin-tone modifiers. The official notebooks skip NFC; on decomposed (NFD) comments their regex teencode replacement corrupts words (e.g. “cảm” becomes “cảmày”). A unit test covers this case.
2. **Markup and identities:**
 and HTML entities are converted, @mentions become “@user”, URLs become “url”, hashtags lose “#”; text is lower-cased.
3. **Tone-mark placement** with VinAI's old-to-new table (“hòa” → “hoà”, “thủy” → “thuỷ”), the convention of PhoBERT's pre-training data.
4. **Elongation:** letters repeated three or more times are squeezed (“đẹppp” → “đẹp”, “quáaaa” → “quá”); “kkk” → “haha”; punctuation and emoticon runs are collapsed (“:)))))” → “:))”).

5. **Emoji:** either kept as separate de-duplicated tokens, or translated into Vietnamese descriptions (😭 → “khóc lớn”) using the paper's dictionary, our additions and the Unicode CLDR Vietnamese names as a fallback.
6. **Teencode:** token-level lookup in 441 official plus 62 added entries (“ko” → “không”, “đc” → “được”).
7. **Word segmentation** (PhoBERT, BiLSTM, TF-IDF word n-grams) with VnCoreNLP RDRSegmenter, the segmenter used for PhoBERT's pre-training (the paper used pyvi); emoji and emoticon tokens bypass the segmenter.

Table 11. Model-specific preprocessing.

Model	Emoji	Segmentation	Tokeniser / vocabulary
TF-IDF + OvR LR	kept as tokens	VnCoreNLP	word 1–2-grams + char_wb 1–5-grams
BiLSTM	kept as tokens	VnCoreNLP	word vocabulary fitted on train (min_freq 2)
PhoBERT-base-v2	→ Vietnamese text	VnCoreNLP	PhoBERT BPE (64k)
ViSoBERT / ViSoBERT-14GB	kept (in vocabulary)	none (syllables)	SentencePiece (15k, includes emoji)

6. Models

M1 — TF-IDF + One-vs-Rest Logistic Regression. Word 1–2-grams on segmented text and character 1–5-grams (char_wb) with sublinear TF and min_df = 2 are concatenated (76,296 features). Each of the 28 labels has an L2-regularised logistic regression (liblinear). $C \in \{1, 2, 4, 8, 16\}$ and class_weight $\in$ {none, balanced} were selected on validation Macro-F1 at $\tau = 0.5$ (selected: C = 4, class_weight = balanced).

M2 — BiLSTM. 300-d embeddings trained from scratch, spatial dropout 0.25, a 2-layer bidirectional LSTM with 256 units per direction, concatenated additive-attention and max pooling, dropout 0.3 and a linear layer with 28 sigmoid outputs. BCE loss, AdamW with a one-cycle schedule (max lr 2e-3), batch 64, up to 25 epochs with early stopping (patience 6) on validation Macro-F1; best epoch 15.

M3 — PhoBERT. vinai/phobert-base-v2 with the [CLS] representation, dropout 0.1 and a linear head with 28 independent sigmoids, trained with plain binary cross-entropy exactly as in the brief. AdamW, lr 3e-5 for the top layer with layer-wise decay 0.9 towards the embeddings, head lr 1e-3, weight decay 0.01, 10% linear warm-up then linear decay, batch 32, max_len 192, up to 10 epochs (patience 3); the head bias is initialised to the logit of each label's training prior, and batches are length-grouped with dynamic padding. Best epoch 9 by validation Macro-F1 at $\tau = 0.5$.

Additional encoders. ViSoBERT (uitnlp/visobert), pre-trained on Vietnamese social-media text with emoji in its vocabulary, and ViSoBERT-14GB (5CD-AI/visobert-14gb-corpus), continued pre-training on 14 GB of Facebook comments, use the same recipe with raw emoji and no word segmentation (lr 4e-5 and 2e-5; the latter was lowered after the ViSoBERT validation curve over-fitted from epoch 4 — a validation-only decision). The **ensemble** averages the saved probabilities of PhoBERT, ViSoBERT and ViSoBERT-14GB with uniform weights; alternative compositions were compared on validation only (ensemble_composition_val.csv) and the planned three-transformer ensemble was kept because adding TF-IDF changed validation Macro-F1 by only about 0.002.

7. Evaluation protocol

One module (src/vigo/evaluation/metrics.py) scores every model on the same split and writes the same result format. Reported metrics: Macro-F1 and Micro-F1 over all 28 labels (headline, as in the paper), Macro-F1 over the 27 emotions only, weighted and sample-averaged F1, per-label precision/recall/F1/average precision, Hamming loss, label-ranking average precision (LRAP) and macro mean average precision (mAP) computed from continuous probabilities, subset accuracy (secondary), parameter count (all nn.Parameters; 28 × (features + 1) for the linear model) and wall-clock training time including per-epoch validation.

Decoding: the main tables use the fixed threshold $\tau = 0.5$ for all labels. Probabilities are saved before binarisation, so two alternative strategies are fitted on validation only and then frozen: a global threshold τ^* maximising validation Macro-F1, and per-label thresholds τ^*_c maximising each label's validation F1 (grid 0.02–0.98, step 0.01).

Hardware and software: Apple M1 Pro, 32 GB, PyTorch 2.14.0 (MPS backend; the BiLSTM runs on CPU where it is faster), transformers 4.57.6, scikit-learn 1.9.1, Python 3.11.16. Seed 42 for Python, NumPy and PyTorch; exact checkpoint revisions are listed in hardware_and_versions.txt.

8. Results

8.1 Main results

Table 11 reports every required metric on the test set at $\tau = 0.5$. PhoBERT improves over the BiLSTM by 0.117 Macro-F1 and over TF-IDF by 0.087. The TF-IDF model outperforms the BiLSTM: character n-grams are robust to teencode and spelling variation, while the BiLSTM learns its embeddings from only 16.5k comments. Among the additional encoders, ViSoBERT-14GB is the strongest (0.620); the three-transformer ensemble is the best system overall, with the lowest Hamming loss and the highest LRAP and mAP.

Table 12. Test results at $\tau = 0.5$ (all required metrics). Hamming loss: lower is better.

Model	Macro-F1	Micro-F1	Macro-F1 (27 emo.)	Hamming	LRAP	mAP	Subset acc.	Params	Train time
TF-IDF + OvR LR	0.542	0.558	0.554	0.0665	0.690	0.560	0.165	2.1M	6 s
BiLSTM	0.512	0.533	0.523	0.0581	0.668	0.539	0.209	5.2M	21.6 min
PhoBERT-base-v2	0.629	0.646	0.637	0.0472	0.769	0.684	0.294	134.4M	51.0 min
ViSoBERT	0.613	0.632	0.622	0.0483	0.755	0.656	0.287	97.0M	41.1 min
ViSoBERT-14GB	0.620	0.642	0.628	0.0470	0.777	0.678	0.298	97.0M	41.9 min
Ensemble (3)	0.643	0.664	0.652	0.0429	0.793	0.710	0.321	328.4M	134.0 min

Table 13. Validation results at $\tau = 0.5$.

Model	Macro-F1	Micro-F1	Hamming	LRAP	mAP
TF-IDF + OvR LR	0.531	0.550	0.0676	0.683	0.553
BiLSTM	0.519	0.545	0.0572	0.668	0.533
PhoBERT-base-v2	0.632	0.644	0.0473	0.770	0.676
ViSoBERT	0.609	0.623	0.0485	0.745	0.647
ViSoBERT-14GB	0.617	0.633	0.0479	0.763	0.672
Ensemble (3)	0.641	0.656	0.0433	0.786	0.711

8.2 Decoding strategies

Fitting thresholds on validation helps modestly on test: the ensemble improves from 0.643 to 0.653 Macro-F1 with a single global threshold ($\tau^* = 0.32$). Per-label thresholds always look best on validation but generalise worse than the global threshold for the ensemble, ViSoBERT and ViSoBERT-14GB — the validation-to-test gap of per-label tuning (last column) is consistently the largest. This is the central observation behind the proposed Final research question.

Table 14. Three decoding strategies; thresholds fitted on validation and frozen before test.

Model	Strategy	τ	Val Macro-F1	Test Macro-F1	Test Micro-F1	Test Hamming	Gap (val − test)
TF-IDF + OvR LR	fixed	0.50	0.531	0.542	0.558	0.0665	-0.011
TF-IDF + OvR LR	global	0.60	0.540	0.538	0.554	0.0596	0.002
TF-IDF + OvR LR	per-label	per-label	0.555	0.539	0.546	0.0666	0.016
BiLSTM	fixed	0.50	0.519	0.512	0.533	0.0581	0.007
BiLSTM	global	0.31	0.524	0.524	0.541	0.0626	0.000
BiLSTM	per-label	per-label	0.542	0.518	0.534	0.0628	0.024
PhoBERT-base-v2	fixed	0.50	0.632	0.629	0.646	0.0472	0.003
PhoBERT-base-v2	global	0.29	0.639	0.633	0.645	0.0535	0.006
PhoBERT-base-v2	per-label	per-label	0.655	0.634	0.647	0.0509	0.022
ViSoBERT	fixed	0.50	0.609	0.613	0.632	0.0483	-0.004
ViSoBERT	global	0.27	0.617	0.616	0.633	0.0517	0.001
ViSoBERT	per-label	per-label	0.634	0.613	0.625	0.0540	0.022
ViSoBERT-14GB	fixed	0.50	0.617	0.620	0.642	0.0470	-0.003
ViSoBERT-14GB	global	0.38	0.623	0.625	0.645	0.0501	-0.002
ViSoBERT-14GB	per-label	per-label	0.644	0.623	0.642	0.0513	0.021
Ensemble (3)	fixed	0.50	0.641	0.643	0.664	0.0429	-0.002
Ensemble (3)	global	0.32	0.655	0.653	0.669	0.0485	0.002
Ensemble (3)	per-label	per-label	0.669	0.649	0.665	0.0477	0.021

8.3 Comparison with the published baselines

The authors report their test numbers at $\tau = 0.5$ with a positive-weighted BCE loss on the same official split. Our PhoBERT alone (0.629) exceeds the paper's best model, ViSoBERT (0.615), and the paper's best PhoBERT configuration (0.593, emoji converted to text). Our ViSoBERT reproduces the published ViSoBERT result (0.613 vs 0.615). We list the differences in our recipe — NFC and tone normalisation, VnCoreNLP segmentation, extended emoji/teencode dictionaries, plain BCE with prior-initialised bias, layer-wise learning-rate decay at a lower learning rate — but we have not ablated them individually, so we make no claim about which factor is responsible.

Table 15. Test Macro-F1 / Micro-F1 compared with the published baselines (same split).

Model	Decoding	Macro-F1	Micro-F1	Source
PhoBERT-base-v2 (paper, emoji kept)	tau=0.5, pos_weight BCE	0.563	0.574	Hung et al. (EACL 2026) official notebooks
PhoBERT-base-v2 (paper, emoji->text)	tau=0.5, pos_weight BCE	0.593	0.604	Hung et al. (EACL 2026) official notebooks
CafeBERT (paper)	tau=0.5, pos_weight BCE	0.612	0.625	Hung et al. (EACL 2026) official notebooks
ViSoBERT (paper, best)	tau=0.5, pos_weight BCE	0.615	0.635	Hung et al. (EACL 2026) official notebooks
TF-IDF + OvR LR (ours)	tau=0.5	0.542	0.558	this work

Model	Decoding	Macro-F1	Micro-F1	Source
TF-IDF + OvR LR (ours)	per-label tau (val-tuned)	0.539	0.546	this work
BiLSTM (ours)	tau=0.5	0.512	0.533	this work
BiLSTM (ours)	per-label tau (val-tuned)	0.518	0.534	this work
PhoBERT-base-v2 (ours)	tau=0.5	0.629	0.646	this work
PhoBERT-base-v2 (ours)	per-label tau (val-tuned)	0.634	0.647	this work
ViSoBERT (ours)	tau=0.5	0.613	0.632	this work
ViSoBERT (ours)	per-label tau (val-tuned)	0.613	0.625	this work
ViSoBERT-14GB (ours)	tau=0.5	0.620	0.642	this work
ViSoBERT-14GB (ours)	per-label tau (val-tuned)	0.623	0.642	this work
Ensemble (3) (ours)	tau=0.5	0.643	0.664	this work
Ensemble (3) (ours)	per-label tau (val-tuned)	0.649	0.665	this work

8.4 Per-label F1

Table 16. Test F1 per label at τ = 0.5 for all models (28 rows), with training frequency and tier.

Label	Train	Tier	TF-IDF	BiLSTM	PhoBERT	ViSoBERT	ViSo-14GB	Ensemble
amusement	2,868	head	0.729	0.686	0.766	0.741	0.766	0.777
excitement	824	mid	0.459	0.380	0.582	0.503	0.467	0.570
joy	1,614	head	0.552	0.519	0.612	0.613	0.612	0.632
love	1,175	head	0.665	0.674	0.721	0.748	0.732	0.764
desire	762	tail	0.458	0.400	0.509	0.540	0.537	0.556
optimism	1,046	mid	0.626	0.594	0.761	0.677	0.713	0.744
caring	1,062	head	0.577	0.556	0.665	0.689	0.689	0.710
pride	774	mid	0.682	0.649	0.719	0.705	0.724	0.729
admiration	883	mid	0.498	0.443	0.640	0.607	0.640	0.656
gratitude	765	mid	0.846	0.819	0.860	0.864	0.878	0.873
relief	635	tail	0.442	0.504	0.607	0.574	0.588	0.632
approval	917	mid	0.551	0.576	0.634	0.635	0.654	0.670
realization	687	tail	0.377	0.244	0.442	0.452	0.450	0.494
surprise	651	tail	0.464	0.394	0.579	0.560	0.568	0.579
curiosity	840	mid	0.541	0.524	0.663	0.633	0.684	0.703
confusion	676	tail	0.393	0.392	0.465	0.481	0.493	0.510
fear	745	tail	0.683	0.632	0.704	0.700	0.655	0.719
nervousness	728	tail	0.430	0.441	0.570	0.541	0.506	0.536
remorse	707	tail	0.705	0.695	0.743	0.716	0.686	0.744
embarrassment	885	mid	0.769	0.754	0.881	0.841	0.823	0.863
disappointment	1,522	head	0.405	0.359	0.534	0.519	0.479	0.499
sadness	2,785	head	0.675	0.636	0.732	0.719	0.750	0.760
grief	747	tail	0.603	0.641	0.707	0.667	0.688	0.678
disgust	1,206	head	0.438	0.427	0.500	0.481	0.494	0.502
anger	1,589	head	0.554	0.532	0.620	0.613	0.656	0.657
annoyance	2,662	head	0.513	0.465	0.617	0.630	0.616	0.647
disapproval	969	mid	0.335	0.187	0.365	0.342	0.405	0.391
neutral	821	mid	0.206	0.221	0.422	0.382	0.417	0.409

9. Midterm questions

Q1. Which emotion labels have the highest and lowest F1?

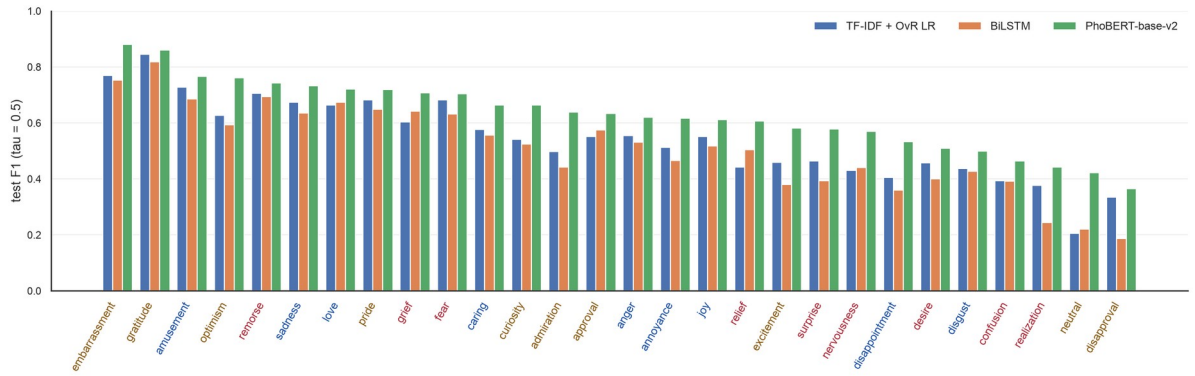


Figure 4. Test F1 per label for the three required models, sorted by PhoBERT (tick colour = tier: blue head, brown mid, red tail).

Answer. For PhoBERT the easiest labels are embarrassment (0.881), gratitude (0.860), amusement (0.766), optimism (0.761), remorse (0.743); the hardest are disapproval (0.365), neutral (0.422), realization (0.442), confusion (0.465), disgust (0.500). The ranking is consistent across the three models: gratitude and embarrassment are near the top for all of them, and disapproval, neutral and realization are at the bottom for all of them — the same weak labels reported in the original paper. The easy labels have explicit lexical markers (“cảm ơn”, “ngại quá”, laughter tokens); the hard ones are implicit and context-dependent (disapproval versus annoyance, a factual realization versus neutral). PhoBERT's largest gains over the baselines are on exactly these context-dependent labels (realization, neutral, disapproval, admiration).

Q2. How strongly does performance correlate with training label frequency?

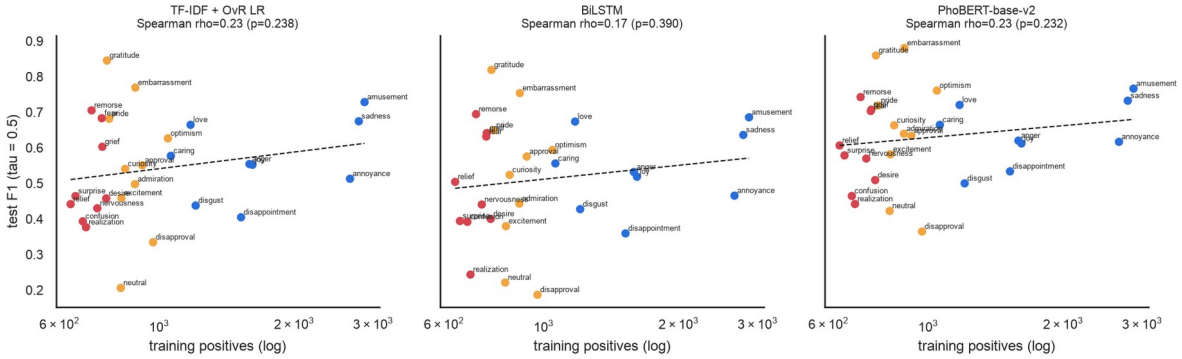


Figure 5. Per-label test F1 against training positives (log scale), with a log-linear fit.

Table 17. Correlation between training frequency and per-label test F1, and mean F1 per tier.

Model	Pearson r	p	Spearman ρ	p	Head F1	Mid F1	Tail F1
TF-IDF + OvR LR	0.221	0.258	0.230	0.238	0.567	0.551	0.506
BiLSTM	0.172	0.383	0.169	0.390	0.539	0.515	0.482
PhoBERT-base-v2	0.186	0.344	0.233	0.232	0.641	0.653	0.592

Answer. The correlation is positive but weak and not statistically significant for any model (Spearman ρ = 0.230, 0.169 and 0.233; all p > 0.2 over 28 labels). Tail labels are on average about five F1 points below head labels for PhoBERT (0.592 vs 0.641), but frequency explains little of the variation: disapproval (969 training positives) is the hardest label, the head label annoyance (2,662) scores only 0.617, while the tail labels remorse and grief exceed 0.70. With an imbalance ratio of only 4.5×, semantic ambiguity and overlap within emotion families matter more than rarity. These are correlations and do not establish causation.

Q3. Which emotion pairs co-occur most, and does the baseline preserve these dependencies?

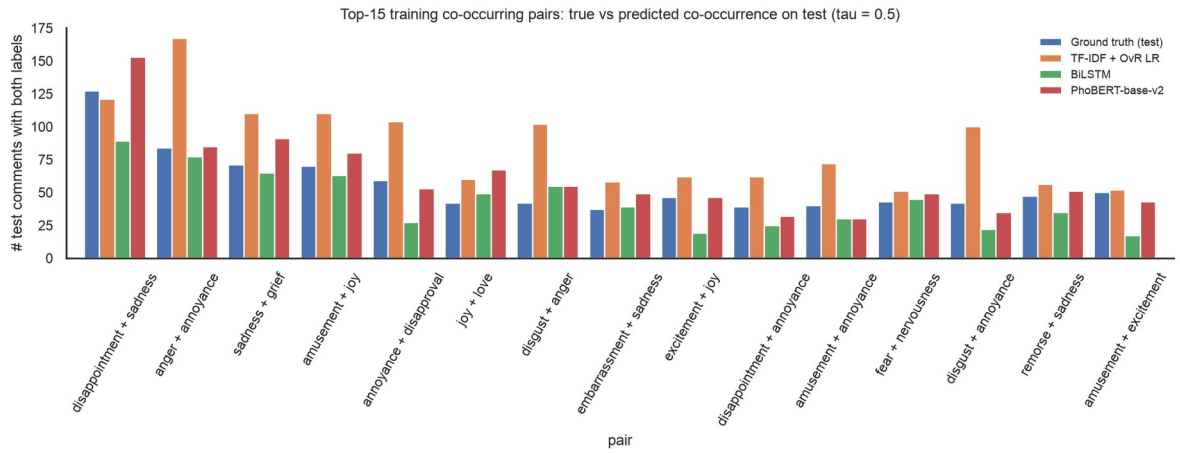


Figure 6. For the 15 most frequent training pairs: true co-occurrence on test versus predicted co-occurrence (τ = 0.5).

Table 18. Global dependency preservation on test (378 label pairs).

Model	r (pair counts)	r (phi)	ρ (phi)	True pairs	Pred. pairs	True |Y|	Pred. |Y|
TF-IDF + OvR LR	0.921	0.918	0.844	2,466	4,698	1.907	2.304
BiLSTM	0.940	0.931	0.887	2,466	1,658	1.907	1.577
PhoBERT-base-v2	0.972	0.962	0.915	2,466	2,314	1.907	1.826

Answer. The most frequent pairs are disappointment + sadness, anger + annoyance and sadness + grief (Table 6). All models are label-independent by construction, so dependencies can only be learned through the shared representation. PhoBERT preserves them best: the Pearson correlation between true and predicted pair counts over all 378 pairs is 0.972, and it predicts 1.826 labels per comment against 1.907 in the ground truth. TF-IDF with balanced class weights over-predicts (2.304 labels per comment, 4,698 predicted pairs versus 2,466 true ones), inflating hostile pairs such as disgust + anger (2.4×). The BiLSTM under-predicts (1.577 labels per comment) and loses pairs such as excitement + joy (0.41×). PhoBERT's remaining distortion is over-predicting joy + love (1.6×) and disappointment + sadness (1.2×).

Q4. On which text styles does PhoBERT most outperform the recurrent baseline?

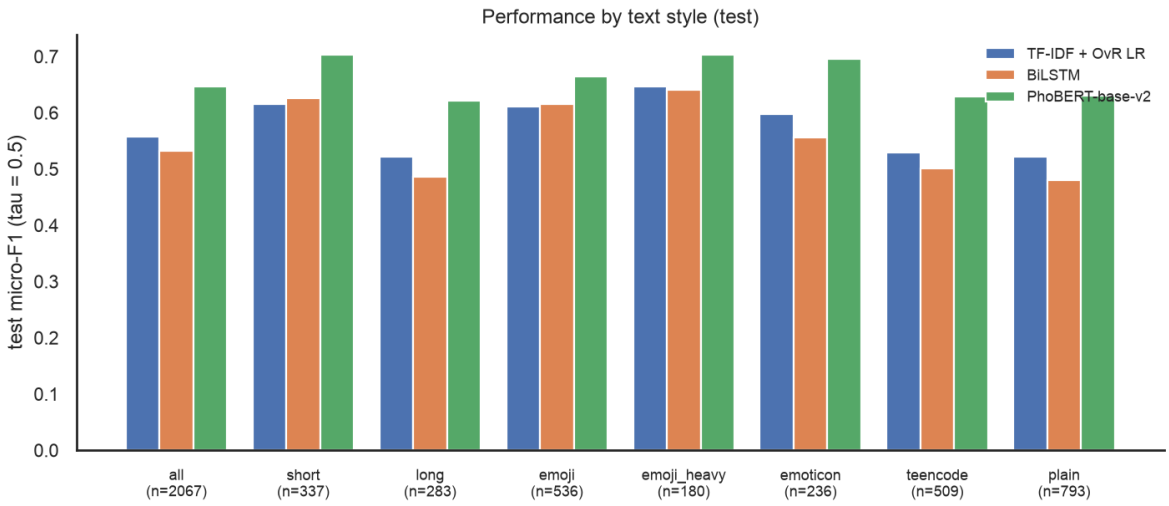


Figure 7. Test Micro-F1 by text-style subset (rules frozen in src/vigo/analysis/text_styles.py before evaluation).

Table 19. Macro-F1 by text style; Δ = PhoBERT − BiLSTM; 95% CI from a paired bootstrap (500 resamples) of the Micro-F1 difference.

Subset	Rule	n	TF-IDF	BiLSTM	PhoBERT	Δ Macro	Δ Micro 95% CI
all	all test comments	2,067	0.542	0.512	0.629	0.117	[+0.098, +0.128]
short	<= 5 whitespace tokens	337	0.575	0.553	0.658	0.104	[+0.042, +0.108]
long	>= 30 whitespace tokens	283	0.473	0.414	0.577	0.163	[+0.099, +0.172]
emoji	>= 1 emoji	536	0.544	0.531	0.590	0.059	[+0.019, +0.077]
emoji_heavy	>= 2 emoji characters	180	0.557	0.531	0.619	0.088	[+0.024, +0.105]
emoticon	>= 1 ASCII emoticon	236	0.493	0.419	0.608	0.189	[+0.099, +0.179]
teencode	>= 1 multi-character teencode-dictionary token	509	0.501	0.468	0.596	0.128	[+0.099, +0.155]
plain	not short, no emoji/emoticon/teencode	793	0.526	0.486	0.628	0.142	[+0.128, +0.174]

Answer. PhoBERT outperforms the BiLSTM on every subset, and every bootstrap interval excludes zero. The largest Macro-F1 gains are on comments with ASCII emoticons (+0.189), long comments (+0.163) and teencode comments (+0.128) — styles where the emotion must be inferred from context rather than a single cue. The smallest gains are on emoji comments (+0.059) and short comments (+0.104): an emoji is a strong lexical cue that the BiLSTM learns directly as a token, while PhoBERT only sees its Vietnamese description. This suggests complementing PhoBERT with an emoji-aware encoder (ViSoBERT), which the ensemble does.

Q5. How sensitive are the outputs to the default threshold of 0.5?

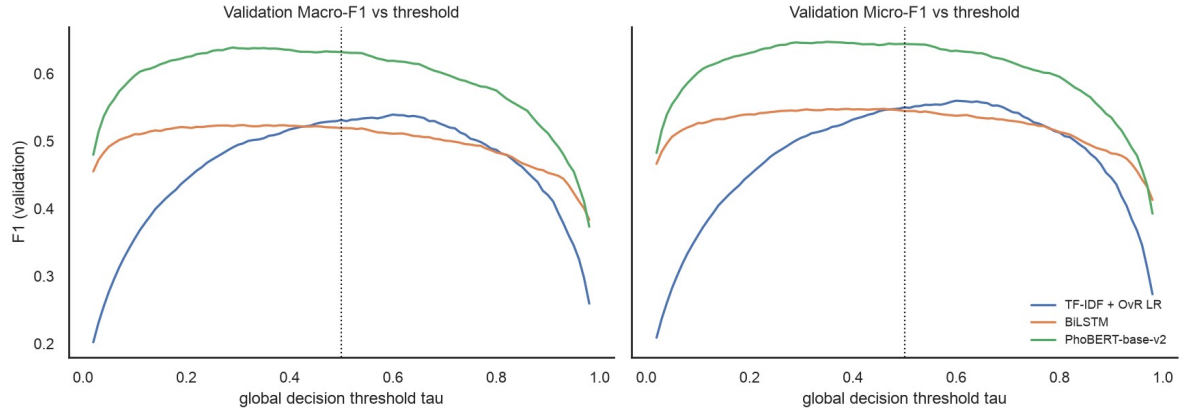


Figure 8. Validation Macro-F1 and Micro-F1 as a function of a global threshold.

Table 20. Validation Macro-F1 at several global thresholds.

Model	τ=0.3	τ=0.4	τ=0.5	τ=0.6	τ=0.7	Best τ	F1 at best	Range in [0.3, 0.7]
TF-IDF + OvR LR	0.495	0.517	0.531	0.540	0.523	0.6	0.540	0.045
BiLSTM	0.523	0.523	0.519	0.511	0.501	0.31	0.524	0.023
PhoBERT-base-v2	0.638	0.635	0.632	0.619	0.600	0.29	0.639	0.038
ViSoBERT	0.617	0.612	0.609	0.604	0.599	0.27	0.617	0.019
ViSoBERT-14GB	0.621	0.623	0.617	0.606	0.585	0.38	0.623	0.038
Ensemble (3)	0.654	0.652	0.641	0.622	0.586	0.32	0.655	0.069

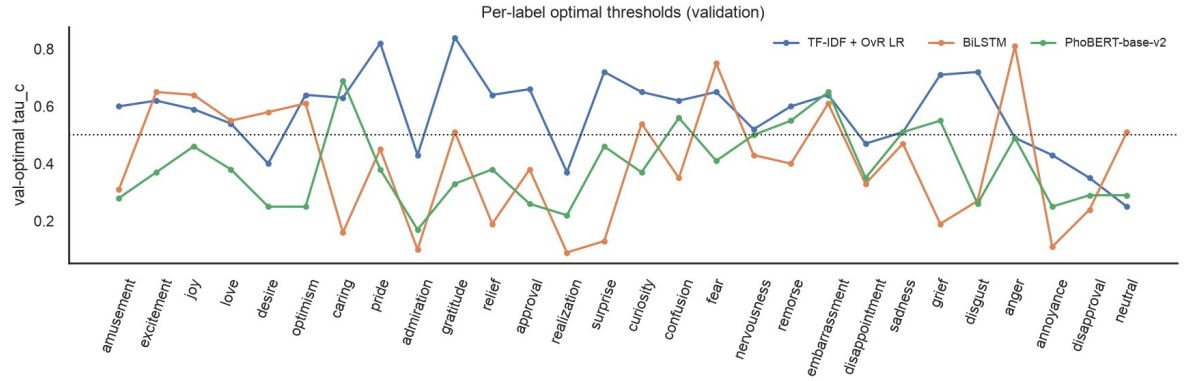


Figure 9. Validation-optimal threshold per label for the three required models.

Answer. On aggregate, 0.5 is close to optimal for PhoBERT: its validation Macro-F1 curve is flat between about 0.25 and 0.5 (best 0.639 at τ = 0.29 vs 0.632 at 0.5) and falls quickly above 0.6. The baselines behave differently: TF-IDF with balanced class weights prefers τ ≈ 0.6, the BiLSTM τ ≈ 0.31. Per label, however, the optimum varies widely — for PhoBERT from 0.17 (admiration) to 0.69 (caring), median 0.38 — so a single uniform threshold is not what individual labels prefer. Per-label tuning adds 0.023 Macro-F1 on validation but only 0.004 on test, a sign of threshold over-fitting on the small validation set.

10. Error analysis

We analyse PhoBERT on the test set at τ = 0.5 (error_by_label.csv, error_confusion_pairs.csv, error_by_bucket.csv). The code also extracts 97 cases for manual audit (25 false negatives, 25 false positives, 15 high-cardinality partial matches, and 8 each of short, emoji-heavy, teencode and emoticon comments); the public file lists ids, labels and probabilities only. Four recurring failure modes emerge:

- **Confusion inside an emotion family.** The most frequent pattern is a missed annoyance while sadness (24) or disappointment is predicted, followed by disgust ↔ annoyance ↔ anger. These pairs belong to the same families found in the correlation analysis (Section 3.3).
- **Implicit, context-bound labels.** Miss rates are highest for disapproval (68.8%), neutral (61.2%), realization (60.0%) and confusion (57.1%); realization is often predicted as neutral and neutral as annoyance.
- **Pull towards about two labels.** Single-label comments receive 0.760 spurious labels on average, while comments with four or more labels lose 1.894 labels; exact match falls from 40.9% (|Y| = 1) to 9.1% (|Y| ≥ 4).
- **Long comments.** Exact match is 19.8% on long comments versus 44.8% on short ones, with 0.887 vs 0.451 missed labels per comment.

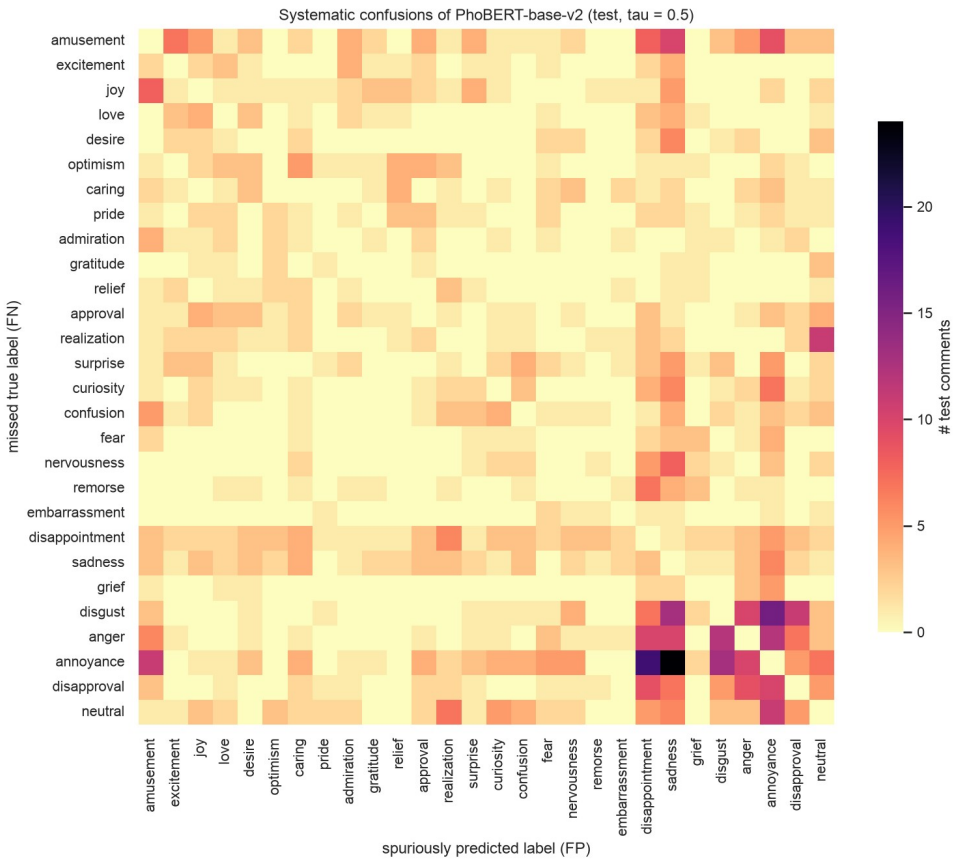


Figure 10. Missed true label (rows) versus spuriously predicted label (columns) in the same test comment, PhoBERT.

Table 21. Labels with the most false negatives (PhoBERT, test).

Label	Support	FN	FP	Miss rate	FP per 100
annoyance	343	145	101	42.3%	4.89
amusement	374	105	59	28.1%	2.85
disappointment	197	90	97	45.7%	4.69

Label	Support	FN	FP	Miss rate	FP per 100
joy	204	80	77	39.2%	3.73
anger	189	78	58	41.3%	2.81
disapproval	112	77	45	68.8%	2.18
disgust	141	75	57	53.2%	2.76
neutral	116	71	52	61.2%	2.52
sadness	345	70	131	20.3%	6.34
realization	95	57	39	60.0%	1.89

Table 22. Error profile by text style and by label cardinality (PhoBERT, test).

Bucket	n	Exact match	Samples-F1	Micro-F1	FN / comment	FP / comment
all	2,067	0.294	0.628	0.646	0.701	0.620
style=short	337	0.448	0.700	0.703	0.451	0.490
style=long	283	0.198	0.600	0.622	0.887	0.767
style=emoji	536	0.308	0.663	0.665	0.660	0.659
style=emoji_heavy	180	0.333	0.702	0.704	0.611	0.622
style=emoticon	236	0.305	0.674	0.696	0.669	0.521
style=teencode	509	0.267	0.610	0.629	0.731	0.699
style=plain	793	0.259	0.602	0.631	0.749	0.618
|Y|=1	695	0.409	0.592	0.570	0.299	0.760
|Y|=2	946	0.276	0.637	0.656	0.728	0.604
|Y|=3	360	0.156	0.668	0.689	1.186	0.450
|Y|>=4	66	0.091	0.645	0.673	1.894	0.318

Table 23. Example audited cases (public version: ids and labels only).

Type	Sample id	True labels	Predicted	Top probabilities
false negative	3919	confusion, fear	(none)	neutral(0.41), curiosity(0.27), confusion(0.13), sadness(0.09)
false negative	15999	amusement, optimism	(none)	admiration(0.36), annoyance(0.30), disgust(0.25), anger(0.22)
false negative	tik005017	caring	anger, annoyance	anger(0.94), annoyance(0.69), disgust(0.15), disapproval(0.01)
false negative	609	amusement, relief	amusement	amusement(0.98), excitement(0.49), curiosity(0.35), joy(0.10)
false negative	682	amusement, neutral	disgust, anger	anger(0.76), disgust(0.62), annoyance(0.50), surprise(0.07)
false negative	12013	amusement	sadness	sadness(1.00), disappointment(0.40), caring(0.09), nervousness(0.02)
false negative	tik013247	disappointment, sadness	realization	realization(0.78), remorse(0.49), confusion(0.44), sadness(0.10)
false negative	20465	approval	nervousness, disappointment, sadness, annoyance	disappointment(0.82), sadness(0.82), annoyance(0.71), nervousness(0.54)
false negative	14149	curiosity, neutral	approval	approval(0.78), caring(0.11), realization(0.09), gratitude(0.04)
false negative	4766	curiosity, disappointment, annoyance	curiosity	curiosity(0.61), amusement(0.36), excitement(0.25), desire(0.04)
false negative	tik012972	remorse, sadness	sadness	sadness(1.00), love(0.33), disappointment(0.29), joy(0.16)
false negative	4040	surprise, disappointment,	disappointment, annoyance	disappointment(0.90), annoyance(0.69), disapproval(0.41), confusion(0.27)

Type	Sample id	True labels	Predicted	Top probabilities
		annoyance		

Implications. The errors point to (i) dependency-aware decoding or modelling for emotion families, (ii) cardinality-aware or per-label decoding, which the threshold study of the Final investigates, and (iii) better handling of long, mixed comments, such as sentence-level aggregation.

11. Proposed Final research question

RQ1 (required): Is a single uniform decision threshold (τ = 0.5) optimal across all 27 emotion labels in the presence of class imbalance?

The Midterm already provides evidence. The per-label optima span 0.17–0.69. Lowering the global threshold raises PhoBERT's Micro-F1 on comments with $|Y| \geq 4$ from 0.673 to 0.750 but lowers it on $|Y| = 1$ from 0.570 to 0.527. And the validation gain of per-label tuning shrinks on test. Planned controlled design:

- use the best Midterm network(s) with the backbone frozen; fit the fixed, global and per-label strategies on validation only and evaluate once on test;
- report Macro-F1, Micro-F1, per-label F1 and Hamming loss, broken down by head / mid / tail tier (rule of Section 3.1) and by $|Y| = 1, 2, 3, \geq 4$;
- report the calibrated threshold profile τ^*_c and the validation-to-test gap ΔF1 to quantify threshold over-fitting;
- robustness: three seeds with bootstrap confidence intervals, a per-label threshold shrunk towards the global one, and a check on the iterative-stratified split;
- optional extensions after RQ1: dependency-aware modelling (RQ2), imbalance-aware losses (RQ3) and noise robustness on teencode and unaccented text (RQ4).

Validation and test probabilities are already stored in the required schema (sample_id, true_label_vector, probability_vector, pred_fixed, pred_global, pred_per_label, label_cardinality) under results/predictions/.

12. Limitations

- All neural results come from a single seed (42); typical seed-to-seed variation for this task is around ±0.5 F1, so small differences (for example between ViSoBERT variants) are not conclusive.
- The recipe differences with the paper were not ablated individually; the gains over the published numbers are therefore not attributed to specific components.
- The official split is only random-level stratified, and the validation set holds as few as 58 positives for the rarest label, which limits threshold estimation.
- The ViSoBERT-14GB learning rate was set after inspecting the ViSoBERT validation curve (a validation-only decision, but still a form of adaptive tuning).
- Teencode dictionaries contain ambiguous entries (single letters such as “t”, “m”), and word segmentation makes errors on non-standard text; neither was evaluated intrinsically.
- CafeBERT and PhoBERT-large configurations are provided but were not run (they need a larger GPU); training times were measured on an Apple M1 Pro and are not comparable with GPU timings.

13. Reproducibility

All results in this report are regenerated from code with bash scripts/run_all.sh (or notebooks/00_colab_quickstart.ipynb on Google Colab), which calls the package command line (python -m vigo eda | split | train-tfidf | train-rnn | train-transformer | ensemble | analyze). The package is organised by responsibility: vigo.data (loading, split, preprocessing), vigo.models, vigo.training, vigo.evaluation, vigo.analysis and vigo.utils. Split ids (splits/), YAML configurations (configs/), saved probabilities (results/runs/, results/predictions/), environment records (hardware_and_versions.txt, results/runs/*/env.json) and unit tests (tests/) are included. Raw comments are not included, in line with the dataset licence.

References

[1] T. Q. Hung, P. T. Nam, S. T. Luu, K. V. Nguyen. ViGoEmotions: A Benchmark Dataset for Fine-grained Emotion Detection on Vietnamese Texts. EACL 2026, pp. 2805–2831. doi:10.18653/v1/2026.eacl-long.129. Code and dictionaries: github.com/ricardo-tran/ViGoEmotions.

[2] D. Q. Nguyen, A. T. Nguyen. PhoBERT: Pre-trained language models for Vietnamese. Findings of EMNLP 2020.

[3] Q.-N. Nguyen et al. ViSoBERT: A Pre-Trained Language Model for Vietnamese Social Media Text Processing. EMNLP 2023; 5CD-AI, visobert-14gb-corpus (Hugging Face).

[4] T. Vu, D. Q. Nguyen, D. Q. Nguyen, M. Dras, M. Johnson. VnCoreNLP: A Vietnamese Natural Language Processing Toolkit. NAACL 2018 (Demonstrations).

[5] VinAI Research. Vietnamese tone normalisation table (BARTpho repository); Unicode CLDR Vietnamese emoji annotations.

[6] D. Demszky et al. GoEmotions: A Dataset of Fine-Grained Emotions. ACL 2020.

[7] A. Graves, J. Schmidhuber. Framewise phoneme classification with bidirectional LSTM. Neural Networks, 2005.

[8] K. Sechidis, G. Tsoumakas, I. Vlahavas. On the stratification of multi-label data. ECML-PKDD 2011.

[9] I. Pillai, G. Fumera, F. Roli. Threshold optimisation for multi-label classifiers. Pattern Recognition 46(7), 2013.

[10] R. Alotaibi, P. Flach. Multi-label thresholding for cost-sensitive classification. Neurocomputing 436, 2021.

[11] C. Sun, X. Qiu, Y. Xu, X. Huang. How to Fine-Tune BERT for Text Classification? CCL 2019 (layer-wise learning-rate decay).

Appendix A. Midterm requirement checklist

Table 24. Brief requirements and the evidence produced by the code.

Requirement	Evidence (results/…)
Sample count and 27(+1)-label inventory	tables/dataset_summary.csv, label_inventory.csv
Label frequency, head / mid / tail	tables/label_frequency.csv, label_tiers.csv, figures/label_frequency.png
Label cardinality distribution	tables/cardinality_summary.csv, figures/cardinality_distribution.png
Co-occurrence and correlation matrix	tables/top_pairs.csv, correlation_matrix_train.csv, figures/cooccurrence.png, correlation.png
Text length (tokens and characters)	tables/text_length_stats.csv, figures/token_length.png, char_length.png

Requirement	Evidence (results/…)
Informal Vietnamese review	tables/informal_text_audit.csv, teencode_top.csv, emoji_top.csv, emoji_rate_by_label.csv
Stratified multi-label split protocol	splits/official/*, tables/split_quality_audit.csv, split_label_coverage.csv
TF-IDF + OvR LR, BiLSTM, PhoBERT (sigmoid + BCE)	runs/tfidf_lr, runs/bilstm, runs/phobert (config, env, summary, probabilities)
Macro/Micro-F1, per-label F1, Hamming, LRAP/mAP, params, time	tables/main_results_test.csv, per_label_f1.csv
Q1–Q5	tables/q1_*–q5_*, figures/q1_*–q5_*
Error analysis on edge cases	tables/error_*.csv, figures/error_confusion_pairs.png
Proposed Final research question	Section 11; predictions/ in the Final schema
Slides, notebooks, README	slides/, notebooks/, README.md